QUESTIONS-GEN — SYSTEM ARCHITECTURE

Qwen3-14B contest problems · src/*.py + questions_gen/ · github.com/chenxingqiang/GenQuestion

DATA

QuestionDataPreparer.py
src/QuestionDataPreparer.py

data_preparer.py
questions_gen/data/data_preparer.py

real_math_data_processor.py
src/real_math_data_processor.py

mixture
historical 50% · variants 30% · novel 20%

THREE-STAGE TRAINER

QuestionsGenTrainer.py
src/QuestionsGenTrainer.py
questions_gen/core/trainer.py
Stage1 SFT Qwen3-14B

reward

RewardCalculator.py
src/RewardCalculator.py reval
0.4 diff 0.3 nov 0.2 rigor 0.1 div

NoveltyConstraint.py
src/NoveltyConstraint.py
GRPO 8-problem groups

DeepSeekTeacher.py
src/DeepSeekTeacher.py
Stage3 distill R1

config.py
src/config.py
questions_gen/core/config.py

VALIDATION questions_gen/validation/

model_validator.py
single-model n tests

batch_validator.py
algebra/calculus/geometry/NT

quality_evaluator.py
content clarity difficulty originality

PACKAGE / CLI / DEPLOY

main_cli.py
questions_gen/cli/main_cli.py
validate batch quality

ollama_manager.py
questions_gen/utils/ollama_manager.py

hf_utils.py
questions_gen/utils/hf_utils.py

questions_gen_training.py
scripts/questions_gen_training.py

ARTIFACTS Hugging Face xingqiang/*

stage1-fp-merged
SFT

stage2-merged-16bit
GRPO

final-merged-16bit
distill quality 0.847

Ollama questions-gen-final
local generate

Stage1 SFT → Stage2 GRPO (RewardCalculator + NoveltyConstraint) → Stage3 DeepSeekTeacher → HF/Ollama